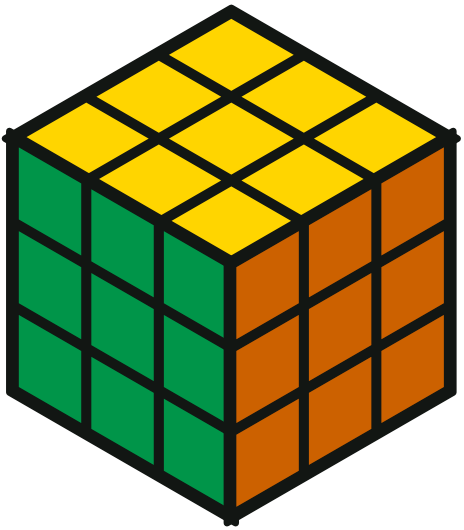
Cube School

Solve the Rubik's Cube.
Step by step.

A little handbook to go with your new cube. No experience needed, just a bit of patience.



To _____

From _____

WELCOME

Enjoy your new cube!

Soon you'll be able to solve it. All the way, all by yourself.

It looks impossible, but it isn't. This handbook breaks the cube into **eight small steps**. Each step has its own goal, and you only need to know one step at a time.

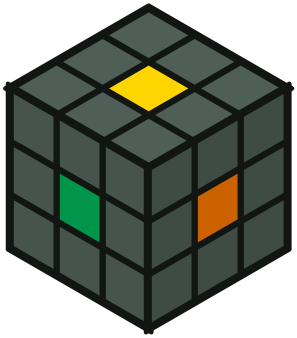
Go at your own pace. It's completely fine if one step takes a whole evening. Turn, look, try again.

1	**The daisy** Four white edges around the yellow centre	☐
2	**The white cross** A white cross underneath the cube	☐
3	**The white corners** The first layer done	☐
4	**The middle layer** Two layers done	☐
5	**The yellow cross** A yellow cross on top	☐
6	**The yellow edges** The edges match the sides	☐
7	**Yellow corners in place** Every corner to its spot	☐
8	**Twist the corners** The cube is solved!	☐

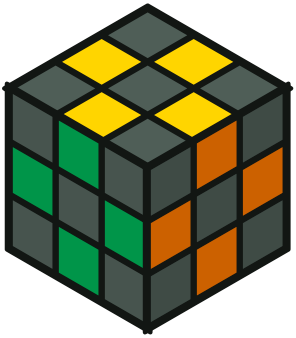
★ Tick the box when you've finished a step. This is the **beginner's method**: the cube is solved layer by layer, with just eight short algorithms to learn.

Meet the cube

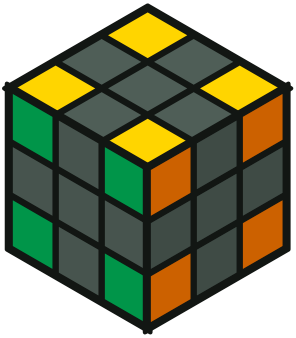
The cube is made of three kinds of pieces.



Centres
Fixed in place, they never move. They decide each side's colour.



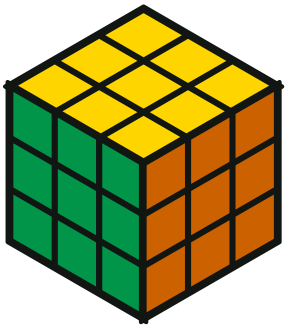
Edges
Two colours. There are 12 of them.



Corners
Three colours. There are 8 of them.

Which colours are opposite each other?

Yellow opposite white. Green opposite blue. Red opposite orange. Check your cube! Different colours? No problem, but the pictures here match a cube in the standard colours best.



Hold the cube like this. Every time!

Yellow centre on top and **green centre facing you**. Then your cube looks just like the pictures in this book.

We build the cube from the bottom up: the white bottom first, the yellow top last.

★ A piece always moves as one whole piece. A corner never turns into an edge, and a centre never changes places.

BEFORE YOU START

The cube's language

Every turn has a letter.



R (Right): right side **up**



R′ (say "R prime"): right side **down**



U (Up): top to the **left**



U′: top to the **right**



F (Front): front **clockwise**, like a clock



F′: front **anticlockwise**



L (Left): left side **down**



L′: left side **up**

A **2** after the letter means: do it twice. **F2** is the same as F F. The arrow under each letter shows which way to turn, looking from the front.

The Dance, the most important move



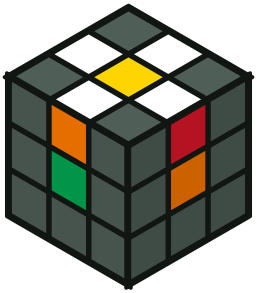
Right up, top left, right down, top right. Practise until your hands do it all by themselves. Fun fact: do the Dance 6 times and the cube is back where it started!

★ Easy to mix up: **R** and **L′** both turn their side **up**. **R′** and **L** both turn their side **down**.

STEP 1 OF 8

The daisy

The yellow centre becomes a flower with four white petals.



The goal: four white edges around the yellow centre, with the white facing up.

The colour on the other side of each petal doesn't matter yet.

Find a white edge and turn it up



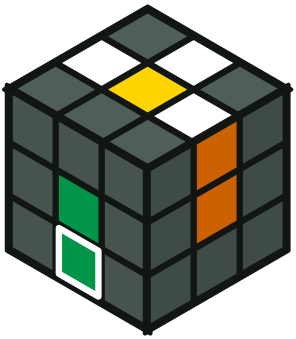
In the middle, white facing you
Turn the right side up.

R



In the middle, white to the side
Turn the front backwards.

F'



On the bottom, white facing down
Turn that side twice.

F2

Edge on the bottom, but the white points to the side? Turn that side once, and the edge moves into the middle layer. Then carry on as above.

A white edge on top, but the white points to the side? Turn it down with the front (F), make room, and turn it back up.

★ **Make room first!** Turn the top (U) so the spot above is empty. Otherwise you'll knock a petal off.

STEP 2 OF 8

The white cross

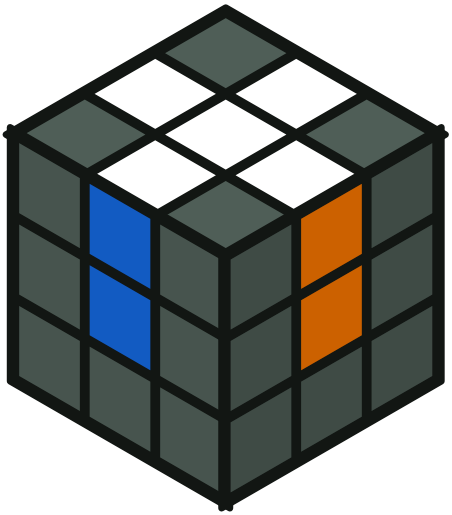
The petals move down one at a time. That makes a white cross underneath the cube.

For each petal

1. Look at the petal's **other colour**, not the white.
2. Turn the top until that colour is above the **centre of the same colour**.
3. Turn that side **twice**. The petal drops down to the bottom.



The green petal is above the green centre. Turn the front twice.



Done! Flip the cube over: a white cross, with its sides matching the centres.

F2 ↻↻

Do the same with all four petals.

★ Match the colour, turn twice. Cross came out wrong? Check that each petal was above the right centre before you turned.

STEP 3 OF 8

The white corners

The four white corners go down between the arms of the cross. Then the first layer is done.



Find the right spot

A corner's two other colours show where it lives. The white, green and orange corner belongs **between the green and orange centres**.

Turn the top until the corner is right above its spot. Hold the cube so the corner is **top right, at the front**.

Dance until it's in!

R ↑ U ← R' ↓ U' →

Do the Dance again and again until the corner is in place with white facing down. Count! Sometimes once is enough, sometimes it takes three or five.

A white corner in the bottom, but twisted the wrong way? Hold it bottom right, at the front, and dance once. It jumps up to the top, and you can start over with it.

★ Done when every side has a little **T** of one colour at the bottom.

STEP 4 OF 8

The middle layer

Four edges go into the middle. Then two layers are done.



Make an upside-down T

Find an edge on top **without yellow**. Turn the top until the edge's front colour is above the centre of the same colour.

The edge's **top colour** shows where it goes. Here it's orange, so it goes to the right, where the orange centre is.

To the right

AWAY DANCE

U ← R ↑ U' → R' ↓

FRONT DANCE

U' → F' ↻ U ← F ↺

To the left

AWAY DANCE

U' → L' ↑ U ← L ↓

FRONT DANCE

U ← F ↺ U' → F' ↻

Both start by turning the top **away** from where the edge is going. Left is the same as right, just mirrored.

An edge in the middle, but flipped the wrong way? Hold it front right and do "To the right" once. It jumps up to the top, and you can put it in the right way.

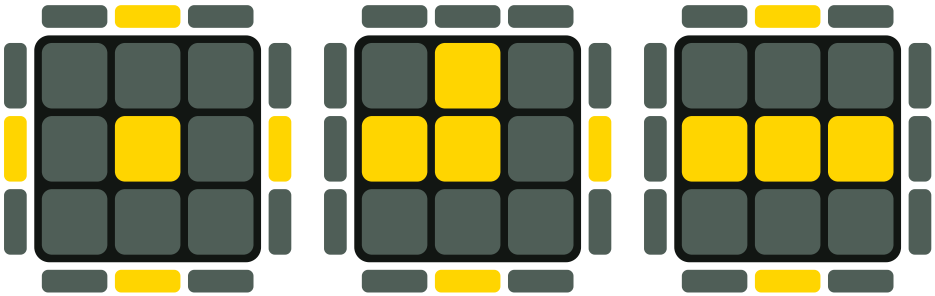
★ Do all the edges on top have yellow? Then an edge is wrong in the middle. Knock it out with "To the right".

STEP 5 OF 8

The yellow cross

Now the top turns yellow. Edges first: a yellow plus sign.

Only look at the plus sign in the middle, not the corners. Your top looks like one of these pictures, seen **from above**:



Dot L Line

Line: hold it sideways, from left to right

FRONT THE DANCE FRONT BACK

F R U R' U' F'

L: one arm to the back, the other to the left

FRONT BACKWARDS DANCE FRONT BACK

F U R U' R' F'

Dot? Do the Line algorithm. Now you have an L. Turn the top so the L sits like in the picture, and do the L algorithm.

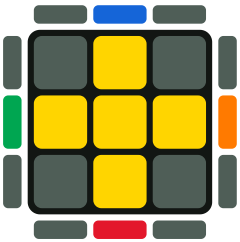
★ The pictures show the top seen from above. The bottom edge of each picture is the side facing you.

STEP 6 OF 8

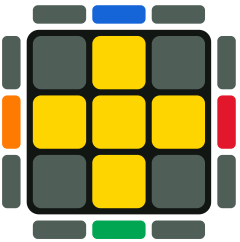
The yellow edges

The cross is yellow. Now the edges need to match the centres on the sides too.

Turn the top until as many edges as possible match the centre below them. At least two will match.



Two side by side match
Here: back and right.



Two opposite each other match
Here: back and front.

Side by side: hold them at the back and right

Do Sune, then turn the top one more step. Now all four match!

FIRST HALF			SECOND HALF				TOP
R ↑	U ←	R' ↓	U ←	R ↑	U2 ←←	R' ↓	U ←

Opposite each other? Do the same algorithm once, from any side. Then turn the top until two edges side by side match, and do as above.

★ Sune mixes up the top corners a bit. Don't worry, we fix those in the last two steps.

STEP 7 OF 8

Yellow corners in place

Now every corner moves to its own spot. They can be twisted the wrong way, we fix that last.



Is the corner in the right spot?

A corner is in the right spot when its three colours match the three centres around it, **even if it's twisted**. Here the front-right corner is in the right spot. The arrows show how the other three move.

Hold the correct corner top right, at the front

FIRST HALF

U ← R ↑ U' → L' ↑

SECOND HALF

U ← R' ↓ U' → L ↓

Look again. Are all four in the right spot? If not, do the algorithm once more, with the same corner still at the front right.

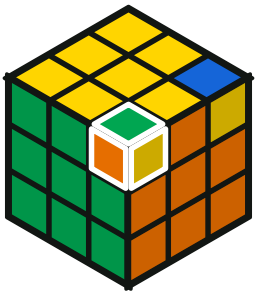
No corner in the right spot? Do the algorithm once from anywhere. Then one corner lands in the right spot, and you carry on as above.

★ The correct corner stays put. The other three go round, one step each time.

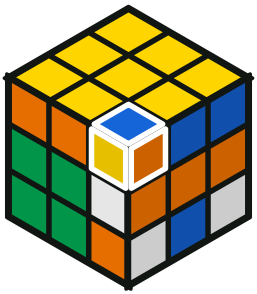
STEP 8 OF 8

Twist the corners

The last step! The corners are in the right spots, now the yellow needs to point up.



The start. Yellow to the right: dance **2 times**.



Then U. The next corner has yellow to the front: dance **4 times**. The bottom looks messy, that's normal.

Hold an unfinished corner top right, at the front

R' ↓ D' ← R ↑ D →

Right down, bottom, right up, bottom back. Do it again and again until the yellow points up.

1. Then turn **only the top** so the next unfinished corner comes to the front right.
2. Dance until the yellow points up. Repeat for every corner.
3. Turn the top into place. Done!

★ **Don't turn the cube around!** Halfway through, the rest of the cube looks messy. That's completely normal. It all comes back together when the last corner is done.

Well done, the cube is solved!

Mix it up and do it again. It gets a little faster every time.

CHEAT SHEET

Every algorithm

Just eight of them. Flip back here whenever you need.

The Dance
Step 3: white corners



To the right
Step 4: middle layer



To the left
Step 4: middle layer



Line
Step 5: yellow cross



L
Step 5: yellow cross



Sune + U
Step 6: matching edges back and right



Yellow corners
Step 7: correct corner front right



Twist the corner
Step 8: 2 or 4 times, don't turn the cube

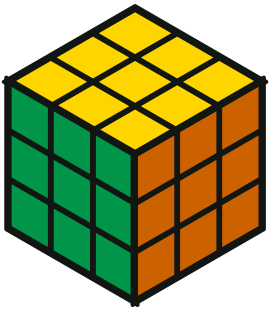


★ Every algorithm is done with the **yellow centre on top** and the **green centre facing you**, unless it says otherwise.

CERTIFICATE

I solved the cube!

All by myself, from mixed up to solved.



Name

Date

My first time

GET FASTER

My times

Write down your times and watch yourself get faster.

DATE	TIME	DATE	TIME

Keep going in Cube School

Watch every step on a spinning cube, follow along move by move on your own cube, practise the algorithms and time yourself. Free, on phone and tablet.

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